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360° View of $\mathbb{R}^d$ Data

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by

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CERTIFICATE

This is to certify that the work presented in this thesis proposal titled ***360° view of*** $\mathbb{R}^d$ ***data*** by *Bipasha Garg* has been carried out under my supervision and has not been submitted elsewhere for a degree.

Date

Advisor: ~~Dr.~~ Kamalakar Karlapalem

Abstract

To be added

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Symbols

α	path loss exponent
B	available bandwidth
β_{AF}	AF gain factor

List of Related Publications

Chapter 1

Related Work

1.1 $\mathbb{R}^d$ Data Visualizations

1.2 Key aspects

1.3 Limitation of the existing work

A Let $x_1 \dots x_n \in \mathbb{R}^d$. Each $x_i = (x_{i1}, \dots x_{id})$, $x_{id} \in \mathbb{R}$. d-dimensional data point. We want to visualize $x_1 \dots x_n$. Now define quadrant, range of value of x_{id} . . . Use example

Chapter 2

Core Visualization

and to comprehend the data.

Visualizing high dimensional data is a challenging task and is crucial in many other domains in the world. The core of this problem is finding a solution inspite all the work that has been done to tackle the problem of cluttering, information loss due to dimensionality reduction, and provide insightful information to the user while taking a note of the cognitive load on the user. A visualization must be intuitive enough for the user to get inferences from it. Our core visualization includes a **360° view of** $\mathbb{R}^d$ **data** along with a **linear matrix-like view** of the distribution of the points in the rings used in the 360° view.

↳ let us check how this develops

2.1 Section 1: Core Idea

A → Not defined.

The core idea of the proposed visualization is to visualize high dimensional data along with providing meaningful inferences to the user. The visualization aims to provide a clutter-free view of the $\mathbb{R}^d$ data, where d is very high dimension, along with a meaningful distribution of the data-points and the labels. The idea of the visualization is to help the user to understand the relative positioning of the data-points and the usage of different quadrants in the space. The visualization focuses on the spread of data-points over various quadrants and takes into account the empty quadrants of the space which are not occupied by any data-point. The visualizations highlights the most used quadrants for the user to understand the distribution of points across the quadrants. The visualization also shows the labels to which the data-point belongs effectively helping the user extract useful information about the data. The main idea also involves the use of 360° of the circle to show the spread of points ~~effectively~~ and the rings used aim to show the hierarchical structure along with showing the deflection of the points while moving from one ring to another.

? without labels data can be visualized

what is useful

2.2 Section 2: Visualization

→ Give range of disk & sectors

what is disk

Our visualization involves the following components:

- **360° Circular Disk Vizualization:** This is the main visualization which contains the data-points to show the arrangement of the points. The Disk is divided into concentric rings which further

↳ what is arrangement

Clarify expand

$d_2 | d_1 | d_5 | d_6 | d_e$

1st (d_2, d_1, d_5) subspace

is this $(d_2, d_1, d_5, d_6, d_e)$ subspace

(p, q)

contains sectors as explained later. The points are plotted in these sectors and connected through edges.

→ Formula $x_i \to (x_{i1} \dots x_{id})$ to (p, q) $p \in$ X-axis $q \in$ Y-axis

- **Concentric Rings:** The disk is divided into concentric rings. These rings represent different **subspaces of the dataset**. These subspaces are chosen by concatenating the features/attribute in the dataset based on increasing variance of the features. As we move outwards from the center the dimensionality of the subspaces increases linearly. Variance is chosen as the metric in order to minimize the spread of the points and get a cleaner view of the positioning of the dataset. The number of rings (subspaces selected) would be equal to the dimensionality of the dataset as there can be a total of 2^d-1, where d is the dimensionality of the dataset, and selecting and plotting all of these would not be computationally feasible for high-dimensional data. It would also lead to miscellaneous information and clutter in the visualization, increasing the cognitive load on the user's brain. The innermost ring represents 1-dimensional subspace and moving outwards the dimensionality increase, the outermost rings representing d-dimensional subspace where d is the dimensionality of the dataset.

What does this mean

Can you improve it.

it should be d right

- **Sector/Quadrant division in each subspace:** Each ring is further divided into 2^d quadrants/sectors which are colored based on what it represents. Each sector represent a bit vector, say 00, 01, 10, 11 etc. which represent the sectors of the second ring which is 2-dimensional. If the last bit of the bit vector 0, which represents negative value, the sector will have a lighter hue as compared to if it is 1 (representing the non negative value).

what about first bit

→ Reorder ? Explain with k-d data quadrants

- **Hierarchical structure:** The rings and the sectors/quadrants represent a hierarchical structure which must be preserved to show accurate positioning of the data-points. Moving from the innermost to outermost ring, the sectors appear to be divided into 2 sectors further depicting negative and non-negative sector in the next ring, just like it is followed in the coordinates axes in space.

why?

Not clear

- **Plotting of data-point:** Each data-point is plotted in this visualization d times, where d is the dimensionality of the dataset. In each ring the data-point is plotted once in its respective sector. The sector is chosen based on the bit vector of the coordinate. Each bit of the data-point's bit vector is plotted in its respective sub-space's sector.

B

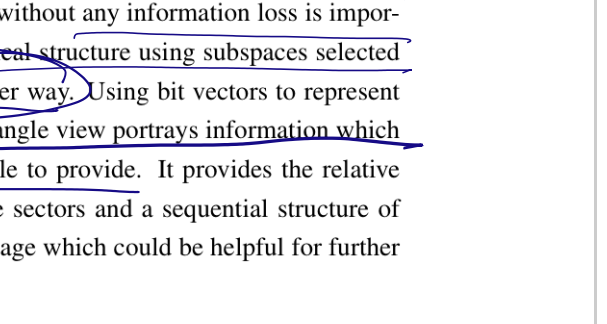
- **Labelling of data-points:** The point labels are shown using different colors, each label having its own assigned color. The data-point (plotted d times in the visualization) is connected using its label's color's edges. Therefore, we can say that one path (connected points) represents one data-point similar to what we see in parallel coordinates and the color represents the label to which the data-point belongs.

data pt (1.2) e

2.3 Section 3: Algorithm

B

Relate need here to algorithm. to clarify



(p,q)
(p,q)
(p,q)
(p,q)
The (p,q)

what are these dim reduction?

$x_i = (x_{i1}, x_{i2}, x_{i3} \dots x_{id})$

$x_{i1}, x_{i2}, \dots x_{il}$ (p, q) pt in 360 view $l < d$

define.

Chapter 3

Relevance of Proposed Visualization

The proposed visualization aims to visualize high-dimensional data in an effective manner which the pre-existing visualization are not able to. The pre-existing visualization as mentioned in *Chapter 1* tend to visualize data either using dimensionality reduction techniques which lead to information or they are not capable of visualizing very high-dimensional data clearly. The proposed visualization is relevant and necessary as the data is growing and there is a need for extracting useful information from the data in the simplest manner.

what is this

The 360° view of the data gives a comprehensive, concise and ~~interpretable~~ view of higher-dimensional data. It provides the user with a clear view of the data by plotting the spread of the data-points across the 360° angle in the disc [refer to section 2.2]. There is need for an intuitive and clear view since reducing dimensions could lead to information loss, therefore visualizing without any information loss is important. Visualizing data using 360° presents the data in a ~~hierarchical~~ structure using subspaces selected based on variance which helps us read the data points in an easier way. Using bit vectors to represent the data makes it easier to plot and read in a visualization. 360° angle view portrays information which the traditional methods like parallel-coordinates etc. are not able to provide. It provides the relative positioning of the data points using hierarchical structure of the sectors and a sequential structure of the subspaces which helps the user to understand the quadrant usage which could be helpful for further study on the data.

What is easier

Need to prove or show

Chapter 4

Closer View - Linear Strip Visualization

Bibliography